

LOSO Experiment B_flex_Istm

LSTM only · Flex only (both hands), no Nothing

Generated 2026-05-11 19:02:11

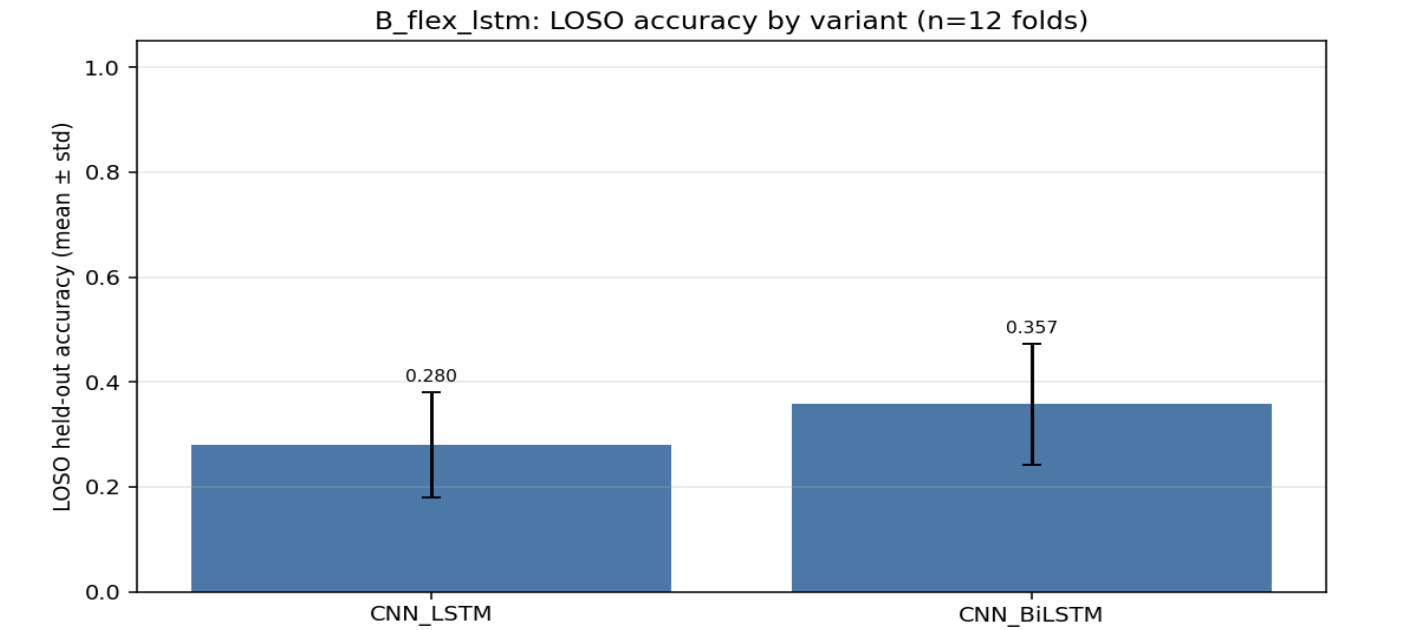
Experiment configuration

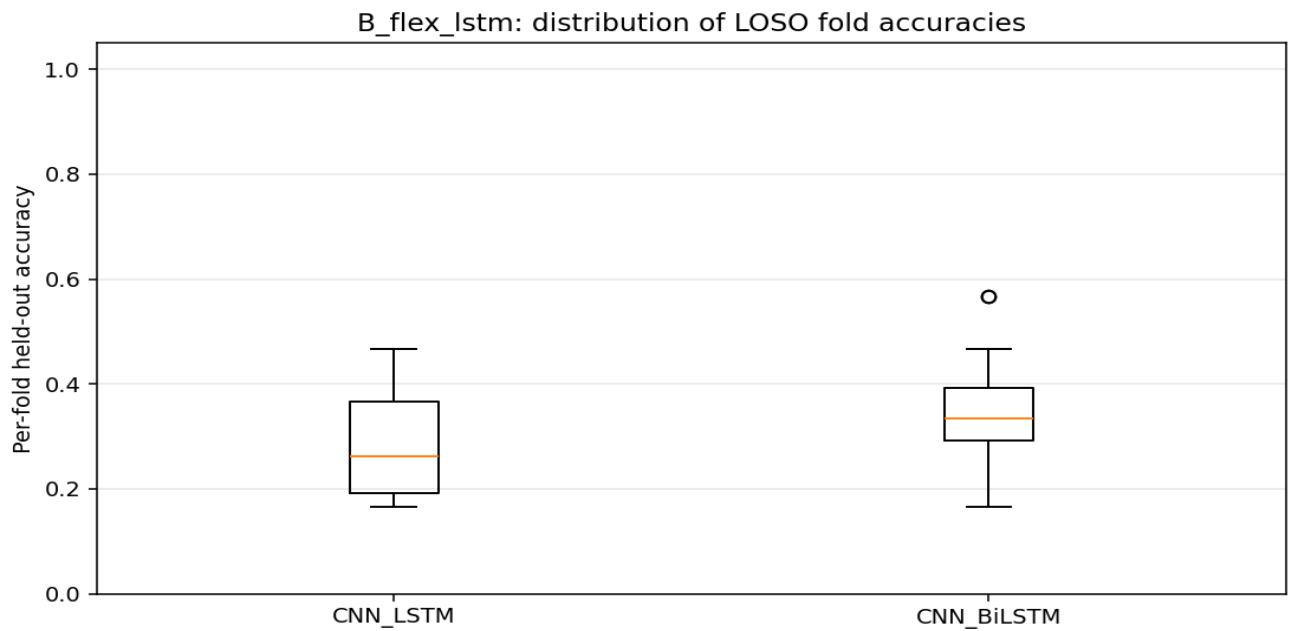
Class set	6 classes excluding Double_Nothing
Classes	Double_Lower, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor modalities	flex
Hands	left, right
Sensor channels	20
Sequence length	90 steps (resampled)
Filter	Butterworth 10.0Hz order 4
Normalisation	minmax
Augmentation	amplitude_scale + baseline_offset, 3 copies
Inner CV folds	3
Epochs · batch size	8 · 16
LOSO subjects	12: 11_Mansh, 12_Marcus, 14_StephenV2, 15_Tash, 16_Bella, 17_Raquel, 18_Bridgette, 1_Alan, 2_Alex, 3_Anghad, 4_Daniel, 6_Harry

LOSO summary across all subjects

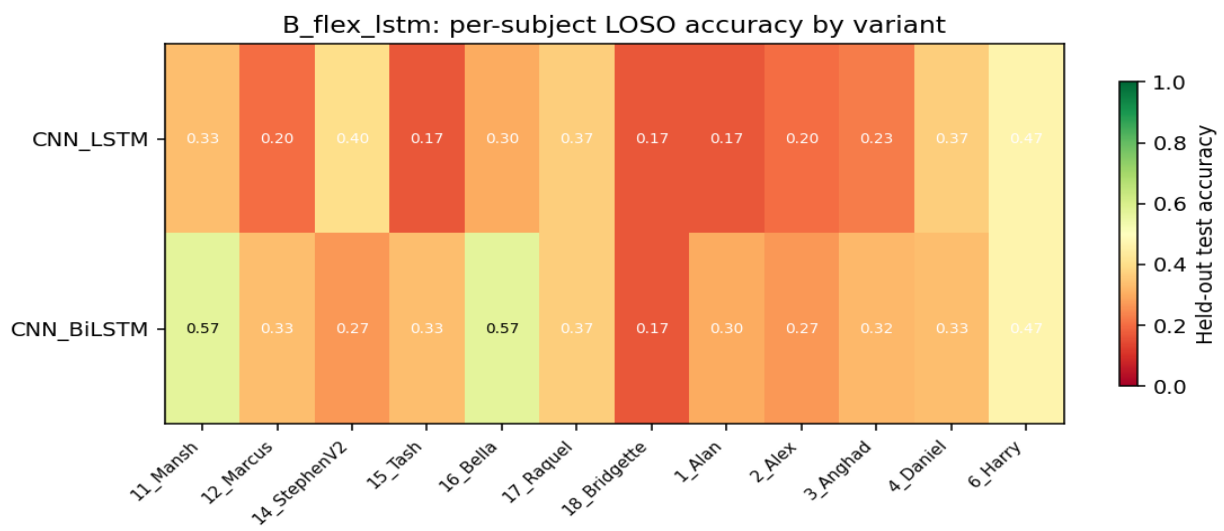
Variant	Params	LOSO mean	LOSO std	Min fold	Max fold	Inner CV mean (avg across folds)
CNN_LSTM	48,422	0.2799	0.1007	0.1667	0.4667	0.3881
CNN_BiLSTM	85,542	0.3574	0.1150	0.1667	0.5667	0.4019

Best variant by LOSO mean: **CNN_BiLSTM**





Per-subject LOSO accuracy heatmap



Per-fold detail (held-out subject = each row)

Held-out subj	Train trials	CNN_LSTM	CNN_BiLSTM
11_Mansh	331	0.333	0.567
12_Marcus	331	0.200	0.333
14_StephenV2	331	0.400	0.267
15_Tash	331	0.167	0.333
16_Bella	331	0.300	0.567
17_Raquel	331	0.367	0.367
18_Bridgette	331	0.167	0.167
1_Alan	331	0.167	0.300
2_Alex	331	0.200	0.267
3_Anghad	330	0.226	0.323
4_Daniel	331	0.367	0.333
6_Harry	331	0.467	0.467

Variant: CNN_LSTM

Params: 48,422

LOSO mean accuracy: 0.2799 ± 0.1007 (min 0.1667, max 0.4667)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	5/5	5/5	0/5	0/5	0/5	0/5	0.333
12_Marcus	5/5	1/5	0/5	0/5	0/5	0/5	0.200
14_StephenV2	5/5	2/5	0/5	0/5	5/5	0/5	0.400
15_Tash	0/5	5/5	0/5	0/5	0/5	0/5	0.167
16_Bella	0/5	4/5	0/5	0/5	5/5	0/5	0.300
17_Raquel	4/5	5/5	0/5	2/5	0/5	0/5	0.367
18_Bridgette	0/5	5/5	0/5	0/5	0/5	0/5	0.167
1_Alan	0/5	0/5	0/5	0/5	0/5	5/5	0.167
2_Alex	0/5	0/5	0/5	5/5	1/5	0/5	0.200
3_Anghad	0/5	0/5	0/5	0/5	4/6	3/5	0.226
4_Daniel	5/5	5/5	0/5	0/5	1/5	0/5	0.367
6_Harry	5/5	4/5	0/5	0/5	5/5	0/5	0.467
MEAN	0.483	0.600	0.000	0.117	0.339	0.133	0.280

Variant: CNN_BiLSTM

Params: 85,542

LOSO mean accuracy: 0.3574 ± 0.1150 (min 0.1667, max 0.5667)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	3/5	1/5	0/5	4/5	5/5	4/5	0.567
12_Marcus	1/5	4/5	0/5	4/5	1/5	0/5	0.333
14_StephenV2	5/5	3/5	0/5	0/5	0/5	0/5	0.267
15_Tash	0/5	5/5	0/5	0/5	1/5	4/5	0.333
16_Bella	4/5	4/5	4/5	0/5	5/5	0/5	0.567
17_Raquel	0/5	5/5	0/5	3/5	0/5	3/5	0.367
18_Bridgette	0/5	5/5	0/5	0/5	0/5	0/5	0.167
1_Alan	5/5	0/5	0/5	0/5	4/5	0/5	0.300
2_Alex	2/5	4/5	0/5	1/5	0/5	1/5	0.267
3_Anghad	5/5	0/5	0/5	0/5	5/6	0/5	0.323
4_Daniel	5/5	0/5	0/5	1/5	4/5	0/5	0.333
6_Harry	5/5	4/5	0/5	0/5	5/5	0/5	0.467
MEAN	0.583	0.583	0.067	0.217	0.486	0.200	0.357